QP Code: 25901042

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU–BVOC (HONOURS) Regular Examination – November 2025

First Semester

Skill Development Components (SDC) – MG1SDCFTA101 – INTRODUCTION TO GOOD FOOD LABORATORY PRACTICES

(2025 Admission Onwards)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A – One or two sentences

Answer any 10 questions, each question carries 1 mark

1. **Identify two protein-rich foods.**
   Meat, Fish, Egg etc.

2. **Expand AOAC.**
   Association of Official Analytical Chemist

3. **Comment on the shelf life of a Certified Reference Material.**
   For some reference standards the shelf life / expiration date may not declared by the reference standards providing organization, in those cases the following shelf life may be considered when the standards are stored un opened at recommended temperature
   - Room temperature items, which are not temperature sensitive and usually are stable for five years from the date of receipt.
   - Refrigerated items usually are stabled for two years from the date of receipt.
   - Freezer items usually are stabled for one year from the date of receipt

4. **List any three documents that must be available at the side of the laboratory equipment.**
   Standard Operating Procedure (SOP), Equipment Logbook / Usage Register, Calibration & Maintenance Records etc.

5. **Expand LOD.**
   Limit of Detection

6. **List two examples of vertical surfaces to be cleaned in a lab.**
   Walls near workbenches or sinks,
   Exterior surfaces of laboratory equipment.

7. **Identify the three basic forms of sample storage in a food lab.**
   There are three basic forms of storage - room temperature (dry room), refrigeration and freezing.

8. **Recognize why lab floors should not be polished.**
   In laboratory environments where water, chemicals, or other liquids may spill, a polished floor can become extremely slippery, posing a major safety hazard to personnel.

9. **Outline the importance of checking expiry dates of chemicals during purchase in a laboratory.**
   Expired or near-expiry chemicals may lose potency or undergo degradation, affecting the accuracy and reproducibility of results.

10. **Define test sample with an example.**
    A test sample is a small, representative portion of a material, product, or substance that is taken for analysis or testing. It is used to evaluate the quality, safety, composition, or performance of the entire batch or lot without testing everything. Any example.

11. **Define an analyst's worksheet.**
    The analyst worksheet provides a written account of the laboratory analytical results. Certain requirements apply to all worksheets.

12. **Define a spiked sample.**
    A sample is split into two sub samples in the laboratory. One is analyzed according to the specified procedure. The other is treated by adding a known amount and concentration of the indicator being measured, running these specified procedures. This should increase the concentration in the spiked sample relative to the un-spiked sample, by a predictable amount. Usually, 10 percent of the sample are split and spiked. They are used to test the accuracy of the laboratory method.

Part B – Short Answer

Answer any 4 questions, each question carries 5 marks

13. **List the duties of support staff in a laboratory.**
    The support staffs of a laboratory is those persons working in and for the laboratory who are not conducting analyses or are not involved in administrative duties. Some examples of duties include:
    1. Glassware washing.
    2. Cleaning and housekeeping maintenance.
    3. Disposal of sample reserves (when no longer required).

4. Pest control.
5. Heavy lifting and moving.

14. Explain why laboratory design should consider long-term work programmes.

Laboratory design must take long-term work programmes into account to ensure that the facility remains functional, safe, and efficient for many years. Key reasons include:

- Future Expansion and Flexibility
- Sustained Safety and Compliance
- Efficient Workflow and Productivity
- Cost-Effectiveness
- Infrastructure Durability
- Technological Upgrades

15. Outline the procedures for environmental control and the key requirements applicable to a food laboratory.

Adequate control of temperature, humidity and dust is important to staff comfort, instrumental performance and safe working (e.g. with flammable solvents). If they are to perform properly optical instruments often require stable temperature conditions. Electronic equipment may have prescribed operating ranges for environmental temperature and humidity. Computers may need to be protected from strong magnetic fields from other equipment; any staff or visitors with heart pace-makers must avoid such fields. Cooling water, either from mains supplies or localised refrigeration may be necessary for the proper functioning of some equipment. Test materials, reagents, standards may need to be stored under controlled conditions. Some substances are affected by sunlight or fluorescent lights and must be protected from it. Delicate balances and optical instruments may need to be protected from vibration (e.g. from blenders, shakers and centrifuges) or may even need stabilised supports. All these needs have to be identified and documented so that proper procedures for monitoring them and taking necessary action can be included in the quality assurance system.

16. List any five hazards that can occur to personnel working in a food laboratory.

- Chemical Hazards – E xposure to acids, bases, solvents, preservatives, or cleaning agents can cause burns, poisoning, respiratory issues, or skin/eye irritation.
- Biological Hazards – Contact with pathogenic microorganisms (bacteria, fungi, viruses) present in food samples can lead to infections or allergic reactions.
- Physical Hazards – Injuries from broken glassware, sharp instruments (scalpels, cutters), hot surfaces, or equipment such as grinders and homogenizers.
- Fire and Electrical Hazards – Faulty wiring, overheating equipment, or flammable chemicals may cause fires, electric shocks, or explosions.
- Ergonomic and Physical Strain Hazards – Repetitive motions (pipetting), improper posture, lifting heavy equipment, or prolonged standing can lead to musculoskeletal disorders

17. Describe the role of management in implementing a QA programme.

Management plays a central and critical role in the successful implementation of a Quality Assurance (QA) programme in a laboratory. Their key responsibilities include:

- Establishing Quality Policies and Objectives
- Providing Resources
- Developing and Approving QA Procedures
- Ensuring Compliance and Monitoring Performance
- Promoting a Quality Culture
- Reviewing and Improving the QA Programme

18. Define proficiency testing and explain its purpose in a laboratory.

Proficiency testing is the part of QA programme, which looks at the accuracy (correctness') of the results actually being reported by the laboratory on real test materials. An independent external assessment of the correctness of the typical result provides an impartial test of analytical quality: this is done by proficiency testing scheme, i.e. methods of checking laboratory testing performance by means of inter laboratory tests. This includes comparison of a laboratory's results at intervals with those of other laboratories. Procedures used when analyzing test materials for such a scheme is those normally used by the laboratory. The proficiency testing is likely to grow in importance because of regional and international developments which require laboratory data to be mutually acceptable between nations for many regulatory purposes.

Part C – Essay

Answer any 2 questions, each question carries 10 marks

19. Explain the importance and objectives of food analysis in the food industry.
Importance of Food Analysis

a) Ensures Food Safety – Detects harmful microorganisms, toxins, contaminants, adulterants, allergens, chemical residues, and spoilage. Helps prevent food-borne illnesses and ensures compliance with food safety regulations (e.g., FSSAI standards).

b) Maintains Quality and Consistency – Monitors sensory, physical, chemical, and microbiological qualities. Ensures that products are uniform in texture, flavor, color, and overall acceptability.

c) Nutritional Labelling and Claims – Provides accurate data for nutrition facts on labels (e.g., protein, fat, vitamins, minerals). Supports health claims, fortification standards, and dietary guidelines.

d) Regulatory Compliance – Ensures that foods meet national and international standards (FSSAI, Codex, ISO). Prevents legal issues related to misbranding, adulteration, and non-compliance.

e) Product Development and Innovation – Helps in formulating new products, improving existing recipes, and optimizing processing conditions. Allows assessment of ingredient functionality and shelf-life.
f) Economic Considerations – Prevents economic fraud by detecting adulteration and ensuring correct ingredient ratios. Ensures fair trade, especially in commodities like milk, oils, spices, honey, etc.

Objectives of Food Analysis
a) Determination of Food Composition – To measure proximate composition (moisture, ash, fat, protein, carbohydrates) and specific components (additives, preservatives, acids, sugars, etc.).
b) Assessment of Food Safety – To detect pathogens, chemical contaminants, pesticide residues, heavy metals, and allergens.
c) Quality Control and Assurance – To evaluate sensory attributes, physical properties (pH, viscosity, texture), and chemical stability. Supports Good Manufacturing Practices (GMP) and HACCP.
d) Verification of Label Claims – To validate claims such as "low fat," "high protein," "sugar-free," "organic," or "fortified." Ensures that consumers are not misled.
e) Shelf-life and Stability Evaluation – To study changes occurring during storage, processing, packaging, and transportation. Helps determine safe and acceptable storage duration.
f) Detection of Adulteration – To identify intentional or accidental adulterants that affect safety, quality, and economic value.
g) Research and Development – To support scientific studies related to food composition, processing effects, nutritional improvement, and novel food design.

20. Explain the importance and methods of maintaining traceability of samples in a food laboratory.

Importance of Traceability in a Food Laboratory

- Ensures Accuracy and Reliability of Results – Traceability allows every sample to be linked to its origin, history, and testing conditions. This helps verify that the final analytical results are correct and scientifically valid.
- Prevents Sample Mix-ups and Contamination – Proper traceability ensures that each sample is uniquely identified, reducing the risk of mislabeling, switching, or contamination—which could lead to incorrect conclusions.
- Supports Quality Assurance and Accreditation – Traceability is a key requirement under ISO/IEC 17025. It demonstrates that the laboratory follows standardized procedures and maintains high-quality operations.
- Facilitates Audits and Regulatory Compliance – Regulatory bodies (FSSAI, BIS, EFSA, FDA) require samples and results to be fully traceable. Clear records help laboratories pass inspections and comply with legal standards.

- Allows Reproducibility and Verification – If results need to be checked, repeated, or verified by another laboratory, traceability provides the necessary documentation to reproduce the testing conditions exactly.
- Supports Investigations and Corrective Actions – In case of a food safety issue or disagreement regarding results, traceability helps identify the source of the problem and implement corrective actions.

Methods of Maintaining Traceability of Samples

- Unique Sample Identification (Sample Coding System)
- Proper Labelling of Containers
- Maintaining Sample Receipt Registers
- Chain-of-Custody Documentation
- Secure and Organized Storage
- Retention and Disposal Records

21. Define calibration. Explain its importance in the accuracy of laboratory results.

Calibration is the process of comparing the measurements of an instrument or equipment with a known, traceable standard in order to determine its accuracy. During calibration, any deviations from the standard are identified, and adjustments or corrections are made to ensure that the instrument provides reliable and accurate results.

Importance of Calibration in the Accuracy of Laboratory Results

- Ensures Accuracy of Results
- Maintains Consistency and Reliability
- Compliance with Standards and Regulations
- Improves Confidence in Test Results
- Prevents Measurement Errors
- Supports Traceability
- Enhances Safety and Quality Control
- Cost Savings and Efficiency

22. Define a Quality Assurance (QA) programme and explain its scope and principles in a food laboratory.

A QA programme may be defined as a mechanism used to ensure that the data are fully reliable, suitable for the intended purpose, presented on time and at an acceptable cost. It is a formal, planned activity whose purpose is to provide assurance that the quality control programme is actually affected and is designed to fit the needs of the laboratory. The scope of the quality assurance system has to be developed in such a way that there is confidence that whenever data are reported.

- the identity and integrity have not been compromised
- the analysis has been conducted by member of the staff who is competent for the task
- the equipments and the methods are appropriate and are working properly
- the laboratory can demonstrate its current capability to produce valid data.

The format adopted in meeting these requirements may vary from laboratory to Laboratory.

Each laboratory (or group of laboratories) meets different requirements, operates within a different organizational environment, experiences different constraints and should have a QA programme, which takes account of these factors.

There are certain factors, which are common

- The use of validated methods
- The use of standard operating procedure (SOP) in the laboratory
- Calibration and traceability of measurement (including use of certified reference material)
- External assessment of performance

While facilities exist for accreditation of laboratories for particular types of work, it is usual to find requirements for these features within this scope of accreditation process along with the requirements for topics which may include organization and management, laboratory accommodation and environment, equipment maintenance, handling of test materials, test methods and quality control procedures (and method performance characteristics), staff training and performance, security, records and reports, sub-contracting of work, outside support services, handling of complaints, quality audit and system review.